Mini Project Report

On

Image Gallery

Submitted To

CODEALPHA

Submitted By

Vedanti S Nasare

Under The Guidance Of

CODEALPHA

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1. Introduction

An Image Gallery is a web-based application that displays a collection of images in an organized and visually appealing manner. It allows users to browse images using navigation controls such as next/previous buttons and lightbox views.

This project demonstrates front-end web development skills using HTML for structure, CSS for styling and layout, and JavaScript for interactivity.

2. Objectives

- Design a responsive image gallery layout using HTML and CSS.
- Implement image navigation using JavaScript.
- Add hover effects and smooth transitions.
- Create a lightbox view for enlarged image display.
- Ensure mobile-friendly responsive design.
- (Bonus) Add image filters/categories.

System Requirements

Hardware:

Computer / Laptop

Internet connection

Software:

Web Browser (Chrome, Edge, Firefox)w

Code Editor (VS Code / Notepad++)

Layout Structure (HTML)

The HTML structure includes:

Header section

Filter buttons (Bonus)

Image grid container

Lightbox modal

Navigation buttons

Styling (CSS)

Features included:

Grid layout

Hover zoom effect

Smooth transitions

Responsive design using media queries

JavaScript Functionality

Features:

Open image in lightbox

Next/Previous navigation

Close lightbox

Output Description

Images displayed in grid format.

Hover zoom effect.

Click opens full-screen lightbox.

Next/Previous navigation buttons.

Smooth transitions.

Mobile-friendly layout.

Category-based filtering (Bonus).

Advantages

User-friendly interface

Responsive design

Smooth animation effects

Easy to customize

Improves web design skills

Limitations

Works only on modern browsers.

Large image files may slow loading.

No backend/database integration.

Conclusion

The Image Gallery project successfully demonstrates how HTML, CSS, and JavaScript can be combined to build an interactive and responsive web application. It enhances understanding of DOM manipulation, CSS Grid, event handling, and responsive web design principles.

This project serves as a strong foundation for developing more advanced front-end applications.